# MONEY

## A HISTORY

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# 1 Mesopotamia, Egypt and Greece

*Then there passed by Midianites, merchant-men, and they drew and lifted up Joseph out of the pit, and sold Joseph to the Ishmaelites for twenty pieces of silver, and they brought Joseph into Egypt.*

GENESIS 37:28 (Authorised Version, 1611)

The earliest-known records of one very important strand in the history of money, broadly characterised by the monetary use of precious metals, can be traced back to the third millennium BC in Mesopotamia and Egypt. Over succeeding millennia this practice has continued, firstly in Europe, the Middle East and South Asia with the spread of coinage, and then throughout the world via the channels of Western colonialism and the rise of modern industrial societies in all parts of the globe. We should not, however, confuse the earliest records of what might appear to be the pre-eminent tradition in the history of money with the origins of money itself.

**6** Ancient Egyptian wall-painting from a tomb at Thebes, *c.* fourteenth century BC, showing gold rings being weighed on a balance. It is unclear whether the Egyptians did in fact use gold rings as money or whether their form on such paintings is simply an artistic convention.

## Mesopotamia and Egypt

We begin, then, with ancient Mesopotamia and Egypt, and by the end of this chapter we shall have arrived at a point in history, about 250 BC, when coinage in gold, silver and bronze was the most widespread form of money throughout most of the Mediterranean world, the Near East and India. While coinage only makes its earliest appearance in the late seventh century BC, the tradition of using precious metals, and especially silver, as money takes us back as far as the twenty-fourth century BC in Mesopotamia.

The transaction envisaged in the passage of Genesis quoted above, an amount of silver in exchange for a slave, is quite plausible for the historical period in which the biblical story of Joseph is supposed to have been set (early second millennium BC), but with one vital qualification. The seventeenth-century English translators of the text wrongly thought that this passage referred to coins ('pieces of silver'). The most important point to remember about the monetary use of metals in Mesopotamia and Egypt is that they passed as bullion by weight

and not as coins, and that consequently the value of the metal to be used in a payment would have been assessed on a pair of scales on each separate occasion. [6] It is also important to place the occurrence of monetary phenomena within the overall context of ancient Near Eastern and Egyptian economic life, which is generally thought to have been typified by a system called 'centralised redistri-

**7** The Ancient Near East and Egypt.

bution', a technical term describing the primary process whereby goods were distributed among the population. This was not done through the action of the market; rather, manufactured goods and agricultural produce seem to have been collected from the people by the authorities, kings and temples, and then redistributed among them according to status and occupation.

Despite the dominance of central power in articulating much of what we would describe as economic life in these ancient societies, silver was recognisably used as money in a number of ways. Several early texts documenting the principles of law and justice in Mesopotamia are preserved on clay tablets and stone inscriptions from the royal and temple archives and monuments of the cities in the region. They provide us with some evidence of the social framework within which precious metal was used as money. In the 'law codes' (which were proclamations of the role of the king in establishing justice rather

than legal codes of practice) a wide range of payments is documented in terms of fixed amounts of weighed silver. Some codes also identify the king as the authority for establishing the standard weights, and some surviving weights also bear the name of the king as testimony to this. Fines for harming another's person or property were usually reckoned in silver. For instance, according to the law code of the king of Eshnunna in northern Mesopotamia (beginning of the second millennium BC), the fine for biting a man's nose was 1 *mina* of silver (about half a kilo), a significant amount, while a slap in the face was reckoned at 10 *shekels*, a sixth as much. The same law code also sets out an ideal price list of common goods, as an expression of the benefits of prosperity brought about by royal justice, listing nine different commodities by weight and volume that should be worth one shekel of silver. Interest rates, too, figure in these quasi-legal texts. Both the Eshnunna code and the more famous Code of Hammurabi of about two centuries later mention an interest rate on loans in silver at 20 per cent. They also show that debts could be paid off in grain with interest according to a silver–grain rate, if the debtor had no silver. Grain by volume seems in appropriate circumstances to have performed monetary functions – for payments to agricultural workers, for instance. The Eshnunna law code quantifies the ideal daily wage of a harvester in both grain and silver (12 *se*, about half a gram). Grain is also used in these texts to express the value of foodstuffs, whereas silver is used to express the value of a wider range of goods, from metals to oil and from lard to wool. From the same period, documents from the city of Ur in southern Mesopotamia containing the balanced accounts of merchants demonstrate that the representation of the monetary use of silver and grain in the law codes largely reflects everyday practice. Many such texts show that silver by weight was used as a standard means of accounting for the value of different goods, and that silver itself was often used as a means of payment in commercial transactions, together with barley.

How did this silver standard work? When the metal itself was paid from one party to another for whatever purpose, silver was weighed out on a balance according to the amount determined by law in the payment of a fine, say, or the sum agreed by the

**8** Stele of the Babylonian king Hammurabi (1792–1750 BC) found at Susa (southern Iran), where it was taken as booty in 1157 BC. This detail of the upper section shows the king, who stands facing the seated sun god with horned headdress. A 'law code' in cuneiform writing, describing many payments in weighed amounts of silver, is inscribed on both sides of the stone and runs to about 3,500 lines. The code continued to be used and copied for over 1,000 years (Paris, Louvre).

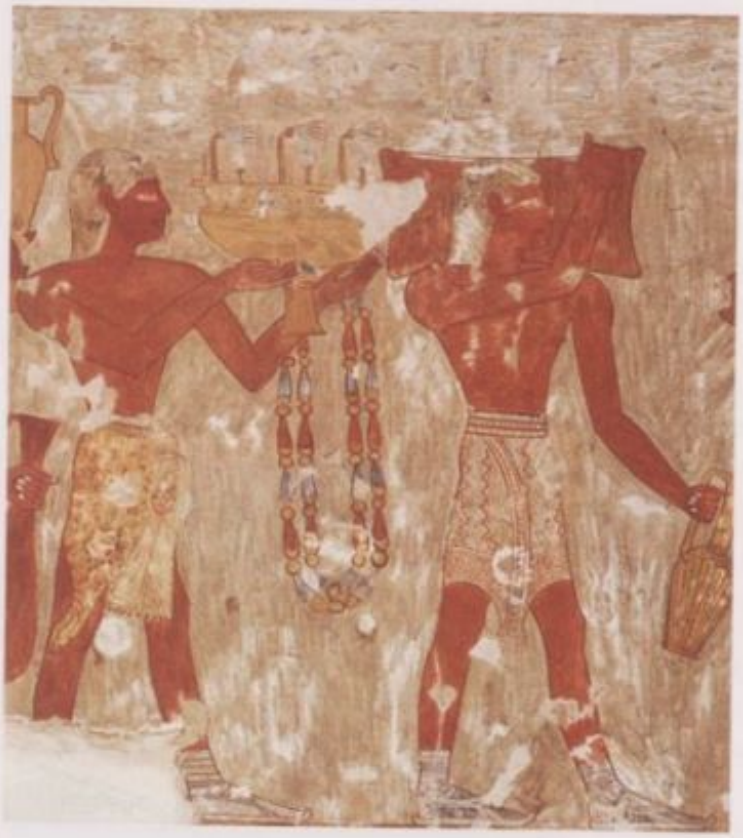
parties to a transaction. Hoards of silver found by archaeologists in Mesopotamia and Iran suggest that the metal was made into large ingots, cut into small scraps or drawn out into thin wire in order to facilitate the correct weighing out of the silver in bullion form. The silver wire was also made into rings which, documentary evidence suggests, were sometimes made to a certain weight.

In Mesopotamian cities, temples played a central role in what we might call the monetary sphere. They were probably the guardians of the official weights, and were generally important in the regulation of the silver system. As prestigious and literate institutions and wealthy repositories of silver and gold, temples were also the centres where records of payments and loans were made and kept. A document dated to 1823 BC (British Museum, WA 82279) records a loan from the temple of the god Shamash in the city of Sippar:

> Puzurum son of Ili-kadari has received from the god Shamash 38⅓ shekels of silver. He will pay interest at the rate set by Shamash. At the time of harvest he will repay the silver and the interest on it.

Silver itself, then, was commonly used as a monetary medium in ancient Mesopotamia. The king and the temples established the weight standards and published in inscriptions the values of certain commodities in silver (though it is uncertain how far this should be understood as representing official price-fixing) as well as the amounts of silver to be paid according to law in a range of circumstances – fines, interest, wages, and so on. But these authorities did not concern themselves further with the public supply of silver. It seems that the monetary 'circulation' of silver bullion was a social custom regulated to some extent by king and temple, but not directly administered by them. The metal had to be imported into the region from surrounding areas, and much of it would have been consumed – in effect hoarded – by kings, aristocrats and temples through taxes, tributes and plunder. As such, silver was a highly valued substance with strong symbolic associations with royalty, wealth and power, and the substantial surplus that was not immobilised in treasuries was potentially available for monetary use.

Similarly deprived of local sources of silver, ancient Egypt was, however, rich in gold from Nubia and in the agricultural wealth resulting from the yearly flooding of the Nile. In a letter to King Amenhotep III (now in the British Museum), Tushatta, King of Mitanni (*c.* 1390–1352 BC), says that in Egypt 'gold was more common than dust'. There is frequent mention in texts from the later New Kingdom (*c.* 1295–1069 BC) of standard weights (the *deben* of 91 grams, and its tenth, the *kite*), according to which metals were used as money in much the same ways as in Mesopotamia – both as a direct means of payment and in accounting for value across a range of goods in transactions by barter. A document preserved from the New Kingdom period records how a policeman, Amunmes,



bought an ox valued at 50 deben of copper (about 4.55 kg) from a workman, Penamun, but only 5 deben were actually paid in copper. The balance was made up by a variety of other commodities whose values were also expressed in deben of copper – fat (30 deben), oil (5 deben) and clothing (10 deben). This text comes from the village of Deir el-Medina near the city of Thebes, inhabited by the skilled craftsmen who worked on the royal tombs in the Valley of the Kings. A large number of such documents have survived from this highly literate community, which show that buying and selling on this system were frequent and that silver was a commonly used metal to express units of value in payments. Indeed, the Egyptian word for silver (*hedj*) seems to have taken on a broader meaning approaching that of money. This can be seen in a document from Deir el-Medina

**9** Wall painting from the tomb of Rekhmira, chief minister to the Pharaoh Thutmose III, fifteenth century BC, showing a man carrying tribute in the form of a copper ingot.

**10** Ancient Egyptian tomb-painting, *c.* fourteenth century BC, showing Nubians from the south of the kingdom carrying tribute in gold rings and other forms.

**11** Part of a hoard of precious-metal objects found at el-Amarna, Egypt. The hoard of gold and silver ingots, wire and fragments was buried in a pot in the fourteenth century BC. (×½)

which records the repayment of a loan of 'silver' (in other words, money) to the sum of 76 deben of copper by one Shedydemduat to a workman named Pennuit. Fifty-four deben had already been repaid, and the remainder still to be returned was therefore 22 deben, 'in order to complete the amount of "silver"'.

Though the amounts are expressed in terms of copper deben, the payments could in fact have been made in a variety of goods. But silver, when available, was clearly highly acceptable as a means of payment, and hoards of gold and silver bullion have been found in Egypt. One such hoard, found during excavations in el-Amarna, the city of King Akhenaten (*c.* 1352–1336 BC), part of which is in the British Museum, contained both gold and silver ingots and pieces of cut silver, much of which is in rings, and a small silver figurine. The weights of some of the individual ingots of gold and silver seem to approximate to multiples and fractions of the deben. The hoard as a whole illustrates the forms in which assorted fragments of precious metal were weighed out and passed from hand to hand in Egypt within a system which commonly used the value of precious metal in payments.

As in Mesopotamia, the evidence for monetary practices from Egypt is of course restricted to the literate elite of the population and those who had access to the services of a scribe. We should perhaps not take their practices as neces-

sarily representative of the life of the people of ancient Egypt as a whole. The artisans of Deir el-Medina were familiar with complex accounting systems, but whether this was also true for the mass of illiterate peasants in their daily lives is doubtful. How far the precious-metal standard extended down the social scale in either Egypt or Mesopotamia may be uncertain in economies which were orientated more towards the redistributive mode than to private enterprise. But as an aspect of the dominant royal, priestly and aristocratic culture, its existence is important for our appreciation of these ancient societies.

## Coinage and bullion

One can observe basic similarities between the otherwise quite different societies of Mesopotamia and Egypt. Common to both is the adoption of a standard of value, expressed either in terms of grain by volume or precious metal by weight. In particular, the weights used in Mesopotamia, the shekel and the mina, became widespread in the Mediterranean world and were adopted by the Greeks in the early first millennium BC. Long-distance trade presumably played a significant role in the use of precious metal as money, because it must have been crucial for merchants to sell their goods in all regions of the Mediterranean. Though most of these would have been exchanged by barter, metals such as gold and silver would also have changed hands. They were valuable possessions in themselves and hence easily valued and exchanged against other goods. The fact that metals were not subject to short-term decay or to yearly fluctuation of supply (like grain) must have increased their usefulness for traders. Thus gold or silver, even in small amounts, could constitute an effective means of making actual payments, whether of a commercial, legal or social kind, because of their general acceptability.

The main restriction on their usefulness was the problem of limited supply,

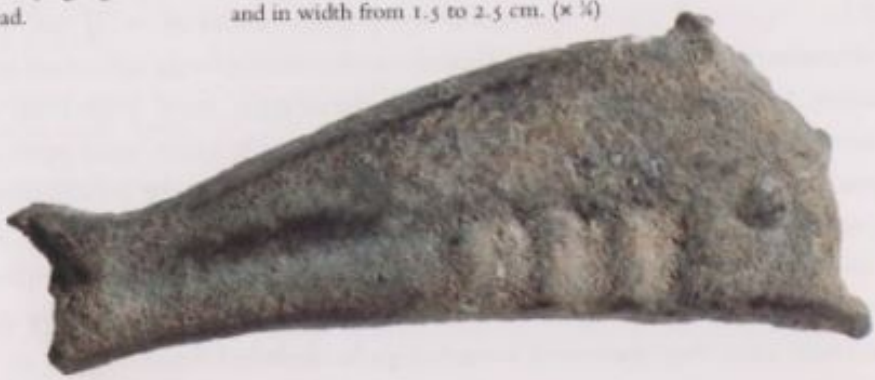
**12** Detail of an obelisk from Nimrud (ancient Calah of the Bible; now in northern Iraq), *c.* 865 BC. The panel shows tribute being paid to the Assyrian king Shalmaneser III in the form of copper ingots, which are being weighed on a pair of scales.

**13** Detail of a bronze strip from the monumental wooden gates of a royal building at Balawat, near Nimrud (northern Iraq), *c.* 845 BC. The scene depicts tribute (perhaps ingots) being brought by the Phoenicians to the Assyrian king.

though this was of course one of the reasons why these metals were valuable in the first place. Regions that found themselves in possession of a source of gold or silver were thus likely to have a privileged access to riches and, often, political power. Such indeed was the case for the kingdom of Lydia in western Asia Minor. The might and wealth of its last king, Croesus (whose reign ended about 547 BC after the conquest of his kingdom by the Persians), was legendary in antiquity. Apart from collecting large sums of money from their subjects, the Lydians were said to derive their wealth from the local river Pactolus and from mines, both of which provided them with electrum, a naturally occurring alloy of gold and silver. From this metal were made the objects that are usually understood as the first coins in the Western tradition, though bronze arrow-shaped objects made from around 600 BC in the area of the Black Sea also 16
seem to represent some form of proto-coinage (coincidentally, the Chinese were also developing 'coinage' in bronze at about the same period). The

**14** Detail of the black obelisk of Shalmaneser III, King of Assyria, found at Nimrud, *c.* 825 BC. The scene shows part of a tribute paid by Jehu, the biblical king of Israel. Among the tribute paid to the king of Assyria and mentioned in the cuneiform inscription above the scene are gold, silver, golden goblets, golden buckets and tin. The figure on the right seems to be carrying ingots above his head.

**15** Bronze or copper bar ingots of the eighth to seventh century BC, found at Tepe Hasanlu in north-western Iran. They range in length from about 20 to 28 cm and in width from 1.5 to 2.5 cm. (× ¼)

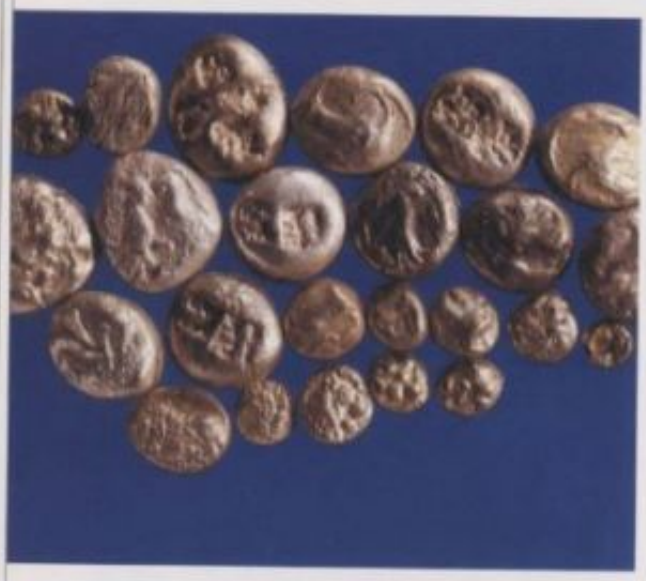


**16** Bronze 'dolphin', made at Olbia on the northern shore of the Black Sea in the fifth century BC. In this area cast bronze objects in the shape of arrow-heads or dolphins seem to have been used as a form of currency.

**17** Electrum coin from Lydia, seventh century BC. The earliest coins were made of electrum, a mixture of gold and silver, and so have a pale yellow colour. Early coins have the imprint of one or more rough punches on one side and a figural design on the other. This coin has the head and neck of a lion, to the left of which can be seen some Lydian letters which represent part of the inscription *Valvel*, a personal name, presumably referring to the person who was responsible for making the coin. (× 2)

Lydian pieces have a rough oval shape, thus resembling nuggets of gold, but they correspond to systems of regular weights, ranging from a group of larger coins (17.2 grams, 16.1 grams, 14.1 grams in weight) down to tiny pieces that weigh one ninety-sixth of the larger pieces. A large proportion of the surviving pieces are of these lesser weights, a fact which points towards a society that was accustomed to precision when it came to dealing with small weights.

The 'coins' are marked with one or more punches on one side, and, although some are blank or have simple parallel lines on the other side, most of them have a figurative design, often an animal (a lion, stag or ram, for example). This large repertoire of types is not fully understood, which makes attribution to specific minting authorities difficult. The lion's head and paws are generally associated with Lydia, as they strongly resemble designs on its later coins. Some designs are associated with cities in Asia Minor (Miletus and Phocaea), though no names are inscribed. There are, however, some rare inscriptions on the electrum which seem to mention individuals rather than states or cities. Two of them, apparently Lydian personal names, read VALVEL and KALIL in Lydian lettering. 17 Another, in Greek, is more explicit and reads 'I am the sign of Phanes' or, on smaller pieces, simply 'of Phanes'. These inscriptions refer to an individual with the Greek name of Phanes, and the grazing stag on the pieces may be his personal badge. Some of the designs on electrum can also be found on seals or rings of the same period, where they definitely represent personal devices, and one may wonder whether individuals were not also responsible for some of the early electrum coins. But who were they? Were they private, wealthy individuals, local rulers or state officials of the kingdom of Lydia? This is an important question for our understanding of the development of coinage, but the question of who made these early electrum pieces must remain open in the present state of our knowledge.



**18** Electrotype copies of electrum coins from the hoard found at Ephesus during the excavations of the Temple of Artemis conducted by the British Museum at the beginning of the twentieth century.

Can we glean any evidence from the places where the electrum coins have been found? The most famous discovery was made during the excavations at the temple of Artemis at Ephesus, where ninety-three electrum pieces and seven unstamped 18 silver nuggets were found underneath the temple itself, suggesting that they were buried there as a religious dedication. Similar coins were probably made on the island of Samos, but in general one can say that the production of these pieces was restricted to western Asia Minor, though some pieces were also found in neighbouring areas (Hellespont, Black Sea). Thus their production and use appear locally rather restricted, a fact that is already suggested by their often minute size: they were not intended to be handled frequently or to circulate

widely in long-distance trade. So why were they made? There is a whole spectrum of non-commercial possibilities: perhaps they were used as a new way of making personalised dedications to the gods, or as prestigious gifts to dependants.

Herodotus mentions that Croesus sent a gift of two *staters* of gold to each of the citizens of the Greek city of Delphi, the site of the famous oracle of the god Apollo, in gratitude for what he took to be a favourable response from the god (*Histories*, 1.54). Though this episode would have happened some time after the first electrum coins appeared, it seems plausible that objects with such attractive designs would have been presented to individuals. No doubt these pieces, being intrinsically valuable, might also change hands in the course of trade, like jewellery or ivory, but that need not mean that they were produced to facilitate such trade in the first instance.

Furthermore, if we look at these early electrum objects from the perspective of the older uses of metal that we met in Mesopotamia, the introduction of a range of stamped designs and fixed weight standards need not appear so very momentous. It is hard to imagine how they could have been used without being weighed on scales, just like pieces of silver bullion, as, from the evidence of the pieces themselves, it is not clear that their value would have been readily intelligible from their size and diameter. This is an important point if the metal in question was to be recognised and accepted at 'face value', that is like later Greek coins, rather than by weight as bullion, as had always been the practice with regard to precious metal previously. But if we look at the pieces from the perspective of the later coinages, to which we will now turn our attention, their appearance in western Asia Minor in the late seventh century BC points towards a crucial change in monetary history.

**19** Electrum stater, perhaps minted during the revolt of the cities of Ionia, Asia Minor, *c.* 495 BC. The Ionian cities, led by Miletus, rebelled against Persian rule from 500 BC. Some electrum coins have been attributed to the period of the revolt, even though silver had by this time become the standard metal for coinage. (× 2)

## The age of silver

Electrum coinage ceased to be minted after the middle of the sixth century, and in its place we find in many regions coinages made of silver and occasionally of gold. A few cities (Cyzicus, Lampsacus, Phocaea and Lesbos) continued to 19
mint electrum coins, but they are very much the exception to the otherwise predominant silver coinage in the late sixth and fifth centuries BC. Why was electrum abandoned? As we have seen, silver had for a long time been the principal metal currency in the ancient Near East, and the succeeding dominance of silver coinage should occasion no surprise. Moreover, being a mixed alloy of two principal metals (but with some other additions such as lead), electrum was not easy to value, as the amount of gold in the alloy could potentially have made a significant difference to the value of a coin.

The earliest silver and gold coins were made in the kingdom of Lydia, 20–21
perhaps in the reign of Croesus (*c.* 560–547 BC). These coins, which are usually

**20** Silver stater from Lydia, Asia Minor, mid-sixth century BC. Some of the earliest silver coins were made in Lydia, with a design showing a lion and a bull. The back of the coin continues to have the crude imprint of two punches.

found in western Asia Minor, depict a forepart of a lion and a bull. Other silver coinages were minted in Caria and other parts of Asia Minor at about the same time as the appearance of the first coins in Greece proper. Aegina (an island off 33
the coast of Attica), Athens and Corinth were the first Greek city-states to mint 34
their own coinages. Although in the earliest stages of silver coinage there are close similarities with the electrum issues in technique of manufacture and design, there are significant differences between the two groups. First, silver coinage was made in a larger geographical area than electrum: it was produced in many places in mainland Greece and particularly Macedonia, as well as in 25
Italy, Sicily, Cyprus, the Aegean islands, Cyrenaica (North Africa), southern 26
France and Asia Minor. Second, it was produced on a much larger scale, and it seems also to have moved into many other regions around the Mediterranean (including those where no coinage was produced, as we shall see below). Third,

**21** Gold stater from Lydia, mid-sixth century BC. Lydian gold and silver coins had the same designs, a lion and a bull. The design was thought to have been introduced by Croesus, the last king of Lydia (*c.* 560–547 BC). The Greeks called the coins 'Croesean staters', although many were minted after the conquest of the Lydian kingdom by the Persians. (× 2)

**22** Scene from the 'Darius vase', made in southern Italy, fourth century BC. The vase is named after the principal figure, who is identified by the inscription as Darius, King of Persia (522–486 BC). It has a scene showing the royal treasurer seated at a table and recording tribute brought to him in sacks. The Greek letters shown on the table represent values from 10,000 shekels to ¼ shekel.

**23, 24** Silver siglos and gold daric of the kingdom of Persia, late sixth to fifth century BC. The 'lion and bull' coins minted by the Persians in western Asia Minor were replaced by coins with new designs, showing the kneeling figure of an archer drawing a

bow (and sometimes holding a spear), although the other side retained the rough punch-marks used on the earlier coins. The silver coins were called *siglos* (the same word as the Hebrew *shekel*), and the gold ones *darics*, named after the Persian king Darius I (522–486 BC), under whom they were first made. (× 1½)

the custom developed of stamping a design or an inscription on the coinage to identify the community that had made it.

The reasons why coinage was adopted are more difficult to understand. Why should some states at certain periods wish to put out silver in this form? Already in antiquity people were discussing this question: in a famous passage of his constitutional treatise, the *Politics*, the fourth-century philosopher Aristotle suggested that the adoption of stamped coins in place of measured pieces of silver was

**25** Silver didrachm of Athens, mid-sixth century BC. Like the earlier electrum coins of Asia Minor and unlike other early silver coinages, the earliest silver coins of Athens were without inscriptions and had varying designs, such as the facing head of a bull. The exact significance of the designs is not clear, though they may have been badges of prominent families or references to the cult of the goddess Athena. (× 1½)

**27, 28** Silver tetradrachms of Athens, fifth century BC and second century BC. From the late sixth century, the various designs of Athenian coins were replaced by the standard design of the head of Athena and her sacred bird, the owl. The design of the earliest pieces was modified in about 480 BC by the addition of olive leaves on Athena's helmet and a small crescent moon behind the owl. The coins were produced in very large numbers from the silver mined at Laurium, and were so important economically that the archaising style was retained throughout the fifth and fourth centuries, although it would by then have come to look very dated. In the second century BC a new design was produced, still showing the owl, but standing on an amphora within a wreath. (× 1½)

simply one of convenience, to save the bother of weighing them out for every transaction. The addition of a design was simply a mark of value (*Politics*, 1257a). Though this explanation sounds plausible enough, it does not answer the question why nobody came up with this solution at an earlier date, if the change from bullion to coinage was so inevitable.

**26** Silver stater of the Aegean island of Naxos, *c.* 500 BC, showing a cup for drinking wine; bunches of grapes are shown on either side and a vine-leaf above. The reverse still has only a punch-mark. (× 1½)

One of the most obvious explanations, one might say, is the minting of silver coins for economic reasons. This was already suggested in the fifth century BC by Herodotus (*Histories*, 1.94), who suggests a link between coins and trade within communities. External trade must also be considered as a possibility. Many early Greek coins have been found in Egypt, the Near East or in the region of the Black Sea, which suggests that the coins were used for long-distance commerce. The silver coins of Athens and northern Greece are frequently found in these areas, and it may well be possible that silver bullion was exported in the form of coinage. In the fourth century merchants coming to Athens could export silver in this form if they did not have any other return cargo of goods. As the writer Xenophon explains in one of his treatises, 'it is sound business to export silver; for wherever they [i.e. the merchants] sell it,

they are sure to make a profit on the capital invested' (*Ways and Means*, 3.2). But the numerous cut coins, ingots or pieces of scrap silver found mixed with coins in hoards indicates that the coins were seen as silver bullion rather than as coins of a fixed value. It is hard to see how long-distance trade alone would have led to the development of coinage, and we need to look at other possible explanations. 29–31

Smaller-scale transactions in communities, involving buying and selling goods for everyday consumption, required small coins. Though low-value coins were used within cities from an early date, as we shall see below, it remains uncertain whether this constituted a vital factor in the production of coinage. The smallest coins, such as half *obols*, would still have been of considerable value and too

**29** Silver stater of the Aegean island of Melos, *c.* 500 BC. The design shows a pomegranate (in Greek, *melon*), a punning reference to the issuing state. This piece was found in a hoard from Asyut in Egypt, where it had been imported from the Aegean along with many hundreds of Greek coins. Like this one, a number of them had been partially chopped up, indicating that they were treated as bullion when they were used in Egypt, an area that did not produce coinage at the time. (× 1½)

**30** Silver stater of Terone, northern Greece, *c.* 500 BC. The design is a wine amphora or storage jar, a reference to the importance of the area for the wine trade. This coin, found at Asyut, had been cut by a chisel, presumably to check that it was made out of solid silver. (× 1½)

**31** Silver stater of Acanthus, northern Greece, *c.* 500 BC. The design on coins of Acanthus is a lion attacking a bull, and the composition has been skilfully adjusted to fit the circular shape. This piece, found with a hoard of Greek coins in the Nile Delta in Egypt, had been tested for quality by a chisel cut. (× 1½)

**32** Scene showing men bargaining with prostitutes, from a Greek painted cup made in Athens, early fifth century BC. The man on the left offers a purse to the woman in front of him. The man in the centre is perhaps trying to fix a price with another woman: he holds up three fingers but she counters with four.

high for everyday retail trade. A member of an Athenian jury, for example, was paid an allowance of three obols a day at the end of the fifth century BC, which probably represented roughly the cost of a whole day's subsistence.

Many coins were issued by Greek cities or rulers, who would have struck coins to meet various payments. As we have seen, Athenian citizens were paid by the state for jury service and for other activities such as attending the Assembly. Military expenditure, such as pay for oarsmen in the fleet or mercenaries, also required coins. It is therefore not surprising to find that states took an interest in minting activities, and that coinage is therefore linked with the increasing role of the state and law in Greek political life. Indeed, states may have made a profit by minting coins: there is some evidence from the Hellenistic period to suggest that a coin had a higher value within its area of issue than the bullion from which it was made. If this were also true in earlier centuries, it could provide a plausible explanation as to why so many Greek cities struck coins and why silver coinage spread so quickly in the late sixth and fifth centuries BC. Considerable savings and revenue would accrue to the states by taking over the production of silver and imposing an overvalued and state-regulated coinage. To work effectively, the system would have needed to fulfil two conditions. First, it would have to be regulated by a legal authority in order to enforce the overvaluation of silver when paid in the form of acceptable coin. The importance of the state's involvement in this regard would explain the choice of designs symbolising its role in maintaining the value of the coin. Second, the state imposing a coinage system would have to have a strong commercial attraction for traders, whether trading in silver itself or in some other commodity, for otherwise they would be put off by the loss involved in accepting overvalued silver coins. This might explain why coinage was produced by those states with a strong 'trade balance' in their favour, such as Athens or 27–8
Aegina. The silver coins of both states became international coinages and are 33
found in many parts of the Mediterranean world.

**33** Silver stater of the island of Aegina, fifth century BC. Aegina was one of the first Greek states to make silver coins. Its early coin design, a turtle, was replaced during the fifth century by a tortoise, as seen here. (× 1½)

We might also look to certain currents in Greek thought for a means of interpreting the rapid and wide spread of coinage as money in the ancient world. Greek philosophers in the sixth and fifth centuries BC were concerned with the concept of law and how it could be applied. Thus the Athenian statesman Solon, who reformed the Athenian laws in the beginning of the sixth century, combines the issues of the city-state, wealth, administration and laws in his writings. The introduction of coinage in Athens took place somewhat later than his reforms, which transformed the Athenian state and society, but the spread of coinage can nevertheless be seen within the general context of the development of the Greek city. One of the Greek words for coinage (*nomisma*) derives from the same etymological root as the word for law (*nomos*), suggesting that the coins were recognised as a product of social conventions. Though ambiguous, our

## Coinage and the city-state

The widespread adoption of silver coinage in the Greek world during the sixth century BC saw the introduction of designs and inscriptions which referred to the authority of the city-states which made the coins. The state's authority guaranteed the quality and value of the coins and protected them against abuses such as forgery.

The designs were chosen to symbolise the authority of the state by referring to its religious cults and mythological past, or by using puns to refer to its name, and were accompanied by inscriptions giving the cities' names. This pattern remained characteristic from the earliest Greek coins to the latest, made five hundred years later in the Hellenistic world, although the shape of the coins and the artistic style changed greatly.

**34a** Silver tetradrachm of Athens, *c.* 510 BC. Athens used the designs of a helmeted head of Athena, the patron deity of the city, and the owl, the bird sacred to Athena; as a result the coins came to be known as 'owls'. Next to the owl are the three Greek letters A, TH and E, the beginning of the Greek word for 'of the Athenians'. The design was introduced in the late sixth century and remained standard until the end of Athenian silver coinage in the first century BC. (× 2)

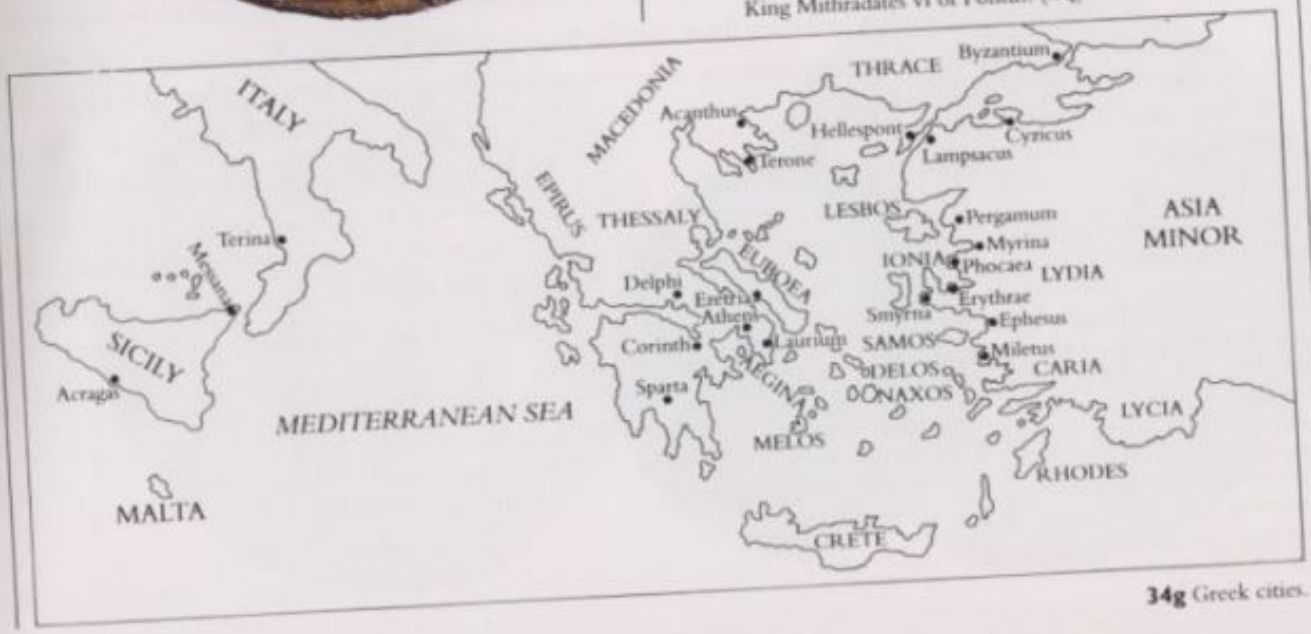
**34b** Silver didrachm of Corinth, *c.* 450 BC. The standard design on coins of Corinth, an important commercial city in southern Greece, was the winged horse Pegasus. According to legend, Pegasus had been tamed by the hero Bellerophon near Corinth. (× 2)

**34c** Silver stater of Tyre in Phoenicia, fourth century BC, showing a bearded figure of the god Melqart riding on a winged sea-horse; beneath the waves is a dolphin. The design reflects the importance of the city as a maritime trading centre. (× 3)

**34d** Silver tetradrachm of Myrina, Asia Minor, mid-second century BC, depicting the head of Apollo, the city's principal deity, wearing a laurel wreath. The much thinner and wider shape of the coin, characteristic of the second century, allowed the engraver greater scope for his treatment of the design. (× 2)

**34e** Silver tetradrachm of Eretria, a city on the Greek island of Euboea, showing the goddess Artemis. Mid-second century BC. (× 2)

**34f** Gold stater of Smyrna, Asia Minor, first century BC. This very rare coin depicts the head of the Tyche or personification of the fortune of the city. The Tyche wears a crown of miniature walls, which are supposed to symbolise those of the city. The reverse represents the cult statue of Aphrodite Stratonikis. The inscriptions refer to the city (*Zmymaion* = 'of Smyrna') and its Executive Board (*prytaneis*), which was responsible for the minting of these emergency gold coins during the war fought in Asia Minor between the Romans and King Mithradates VI of Pontus. (× 4)




**34g** Greek cities.

**35** Silver tetradrachm of the Greek city of Messana (modern Messina), Sicily, fifth century BC. The coin depicts a hare above a cicada. The design was mentioned by Aristotle, who states that Anaxilas, the tyrant of Messana, was responsible for introducing hares into Sicily; this may perhaps suggest that the coins were called 'hares', just as those of Athens were called 'owls'. (× 2)

literary evidence for laws on weights and measures in Athens is particularly rich for the sixth century, and it comes as no surprise that the increased use of silver coinage falls into the same period.

It is clear from the preceding discussion that the origins and spread of silver coinage may have had different sorts of explanations. But once coinage had spread to cities across the Greek world it seems that it rapidly became the predominant form of precious metal money there. There is much evidence for the widespread use of coins from fifth-century Athens in the literature of the period. A number of inventories and accounts of state expenditure (like those of the building works of the Parthenon on the Acropolis) have survived inscribed on stone. It is clear from these inscriptions that coinage was the normal form of money, although uncoined bullion or precious metal in other forms was also mentioned, in particular in the lists of assets belonging to temples. According to the historian Thucydides, the great Athenian statesman Pericles made a speech on the eve of the Peloponnesian War against Sparta (431–404 BC) in which he reminded the Athenians of their financial strength (Thucydides, 2.13). He mentions that their annual income from the allies was 600 talents, while their reserves consisted of

> 6000 talents of coined silver . . . as well as uncoined gold and silver in the form of private and public dedications, the sacred items used in festivals and games, plunder from the Persians and other things, worth no less than 500 talents. There were also substantial sums in other temples, which they could use, and if all else failed they could use the gold which adorned the goddess herself [i.e. the colossal gold and ivory statue of Athena on the Acropolis]. . . . But he said that if they used it for their own preservation then they must restore at least as much.

**36** Silver stater of the Greek city of Terina, southern Italy, fifth century BC. The winged figure on the coins of Terina is probably a personification of the city, but the iconography is complex. Like Victory, she is winged and holds a wreath; she sits on an amphora, perhaps symbolising the source of a local spring or river, and holds a *caduceus*, a symbol of peace. (× 2)

**37** Gold stater of Panticapaeum, a Greek city in the Crimea on the northern coast of the Black Sea, late fourth century BC. The coin has a head of the god Pan, a punning allusion to the city's name. The minting of gold coins, otherwise unusual, reflects the availability of gold in the region, a wealth exemplified by the rich burials of Scythian rulers in this area, which often included gold objects. Pan is shown with characteristically pointed ears and a snub nose, and wearing a wreath of vine-leaves. (× 5)

**38** Silver tetradrachm of Rhodes, early fourth century BC. The island of Rhodes was one of the most prosperous states in the fourth and third centuries BC and issued many coins. The head of Apollo is shown almost full-face, a mark of the skill of the engraver who made the die. The rose on the back refers to Rhodes, since the Greek for rose was *rhodos*. (× 3)

This passage shows the primacy of silver coinage over other forms of wealth, both in quantity and in function: temple treasures and the like could be used *in extremis*, but would have to be replaced later. Coins in silver formed the bulk of the city's wealth. But while silver *tetradrachms* were suitable for state transactions or large payments, the lowest silver denominations in the fifth century were still too valuable to cater for small-scale individual purchases. Nevertheless, people seem to have adapted to using coins, and eventually the smallest of the silver coins were replaced by bronze coinage. The idea of making low-value coins out of bronze, a cheap metal, seems to have originated in the cities of southern Italy in the late fifth century BC, but spread throughout most of the Greek world during the fourth and third centuries BC. From then onwards any transaction in the market-places of Athens or other cities could have been carried out in coinage: the new medium of bronze provided small enough coins, such as the *chalkous* (an eighth of a silver obol). 39–40 41

**39** Bronze 'coin' made at the Greek city of Acragas (modern Agrigento) in Sicily, *c.* mid-fifth century BC. Before the introduction of bronze coins, pyramid-shaped pieces of bronze currency were cast in Sicily. This example is decorated with two eagles' heads and three dots on top to indicate its value. (× 1½)

Coinage is only one part of the story of money in the Greek world, albeit the most significant. Though most ancient cities may have used coins, the majority did not actually produce coinage themselves in the Archaic and Classical periods (*c.* 600–320 BC) for a variety of reasons – some perhaps because they had no access to silver and some for political reasons. The most famous of these cities was Sparta, where no coins were made until the third century BC, and the Spartans may have used iron spits instead. The ownership of gold and silver was regarded as contrary to the Spartan warrior ethos and characteristic of a base, mercantile mentality which the manly Spartans affected to despise. Nevertheless, to defeat the wealthy Athenians in the Peloponnesian War, they had to accept huge subsidies in coin from the Persian king in order to finance a fleet with which to challenge the Athenians' maritime power.

**40** Bronze coin of Acragas, late fifth century BC. After making objects like **39**, the city produced coin-shaped bronzes by the normal coin-striking process. This piece depicts a crab and other sea creatures, reflecting the city's maritime situation. The six dots which surround the design represent its value. (× 1½)

## Money and credit

International warfare faced Sparta with new demands which its primitive monetary system could not meet unaided. By contrast, other states in Greece, in particular Athens, used coins and money in a more developed way, such as through banks. Athenian banks are first attested towards the end of the fifth century BC, but most of our evidence comes from the fourth. We should not think of these as sophisticated financial institutions like modern banks, but as a cross between a *bureau de change* and a pawnbroker. Bankers operated in a private capacity and were not subject to any state regulation. Their most visible activity was money-changing, primarily for foreigners who arrived at Athens with non-Athenian currency. We hear of moneychangers doing business from behind tables set up in the Agora (market-place); even now the modern Greek word *trapeza* means both 'table' and 'bank'. Bankers also accepted money on deposit, but appear to have paid no interest. What they offered was effectively a safe-deposit box – a

service of particular use to foreign merchants who had nowhere else to store their valuables while at Athens. But we also hear of Athenians who kept money with a banker, possibly for security but perhaps in order to conceal the extent of their wealth from the prying eyes of the taxman. Finally, bankers lent money – both their own and that of depositors – to private individuals, typically at an interest rate of 12 per cent a year. People naturally preferred to borrow from friends or relatives, and most credit transactions were of this kind. Bankers therefore tended to be used only as lenders of last resort.

**41** Bronze coin of Athens, late fourth century BC. This, one of the earliest bronze coins made by the city, shows an owl with two bodies, perhaps suggesting a double denomination (2 chalkoi or 2 kollyboi). This type of coin was minted in huge numbers, and almost a thousand specimens have been recovered so far in the excavations in the Agora or central square of Athens. (× 2)

The banker of whom we know most, Pasion, rose from being the slave of a banker to acquire not only control of the bank but also Athenian citizenship. By the time of his death in 370 BC he was worth over 60 talents (360,000 *drachmae*). The scale of his private wealth is graphically illustrated when we recall that in the fifth century the rich city of Byzantium paid 16 talents a year to Athens. But banking was a high-risk business: Pasion was nearly ruined early in his career when he was sued by a disgruntled client, and we hear of many other banks going broke.

In the Greek world there was no equivalent of a state bank. The surplus wealth of Athens was stored under the protection of Athena on the Acropolis and was regarded in some sense as belonging to the goddess, so that thieves would be deterred by fear of committing an act of sacrilege. Hence, when the Athenians needed extra money to finance military operations, they 'borrowed' it from Athena with a promise to repay it when they could. When their finances were in crisis, they even melted down the gold cult-statues of the goddess. We also know that some temples lent their money to individuals. Inscribed accounts of the fifth century from a temple in the Athenian countryside show loans being made in units of 200 and 300 drachmae.

In the Greek world financial transactions were conducted solely in coined money; there were no cheques or exchangeable bills. Therefore the smooth running of the economy required there to be sufficient coins in circulation. In Athens we find that coins were minted on an increasingly regular basis and in

**42** Bronze coin of Ptolemy II, King of Egypt, 286–246 BC. The coinage system set up in the Ptolemaic kingdom of Egypt was designed for the purpose of reserving silver for dealing with foreign merchants. Consequently, large bronze coins were made in substantial quantities to meet the internal needs of the kingdom. The coin depicts a head of the god Zeus Ammon, with a ram's horn. (× 2)

**43** Bronze coin of Agathocles, King of Bactria (modern Afghanistan), early second century BC. Like the Hellenistic kingdoms further west, the kingdom of Bactria produced bronze coins, although these were sometimes square, like Indian coins. On one side the legend was in Greek, but the other, which depicts a figure of an Indian goddess holding a lotus, has an Indian (Brahmi script) inscription.

# Hellenistic portraiture

The vast conquests of Alexander the Great (336–323 BC) transformed the ancient world. New areas, from Greece to India, fell under the domination of Greek rulers, and the Greek world was transformed from a collection of city-states into a series of monarchies, known today as the Hellenistic kingdoms. These changes had profound effects on the coinage. Royal portraiture was introduced shortly after Alexander's death, while the spread of Greek culture to the east meant that similar designs were also adopted there, in the kingdoms which succeeded the empire of Alexander.

Since most rulers wished to emulate Alexander or claimed to be his legitimate successors, their portraits usually show them in his pose, with the head turned upwards and eyes raised, and with the *diadem*, a white ribbon which was the symbol of kingship. The portraits died out in the first century BC, when the independent kingdoms succumbed to the Romans, but although the Romans had a very different tradition of portraiture, the earliest emperors used portraits influenced by those of the Hellenistic kings.

**44c** Silver tetradrachm of Eumenes II, King of Pergamum, Asia Minor, 197–158 BC. The kingdom of Pergamum achieved its most brilliant period in the reign of Eumenes II, whose coins depicted a posthumous portrait of Philetaerus, the founder of the royal dynasty in the third century BC.

**44d** Silver tetradrachm of Antiochus VI, King of Syria, mid-second century BC. The young king is shown wearing the royal diadem and a crown of rays. Antiochus regarded himself as a god on earth and this crown was a symbol of divinity. The use of divine symbols was a feature of the portraits of Hellenistic kings of the late fourth and early third century BC, but is unusual on coins of later kings such as Antiochus VI.

**44a** Silver tetradrachm of Lysimachus, king in Thrace (northern Greece) and Asia Minor, 305–281 BC. The coin depicts Alexander the Great, since Lysimachus wanted to legitimise his position by emphasising that he was Alexander's successor. The portrait has the characteristic hairstyle of Alexander, and, as well as the diadem, his head is decorated with a ram's horn, a symbol of the god Zeus-Ammon, whom Alexander had claimed to be his father.

**44b** Gold octodrachm of Ptolemy II, King of Egypt (286–246 BC), and his second wife Arsinoe. The Greek legend *adelphon* refers to the fact that Ptolemy (shown nearer the viewer, wearing the royal diadem) and his queen were also brother and sister, in continuation of earlier Egyptian royal practice. (× 3)

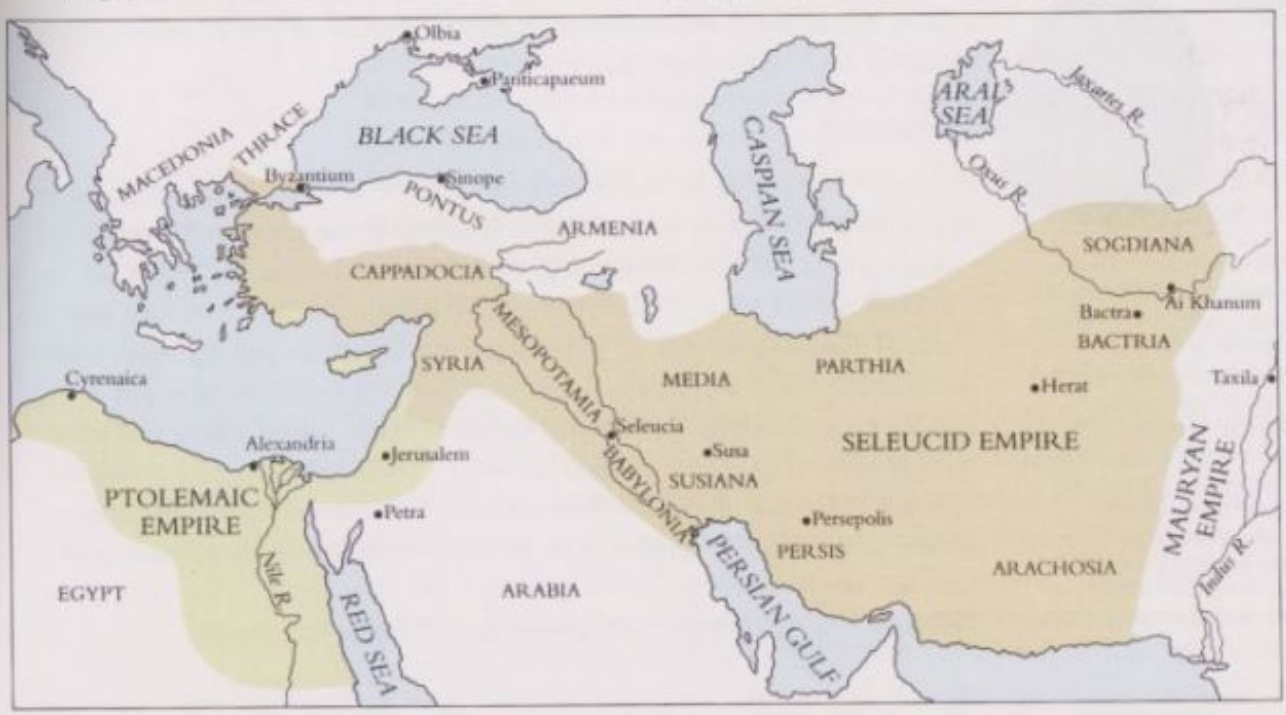
**44e** Silver tetradrachm of Mithridates II, King of Pontus (modern northern Turkey), mid-third century BC. The coin portrait itself and the use of the diadem show how Greek ideas were adopted in other areas by non-Greek dynasties in the wake of Alexander's conquests. (× 3)

**44f** Silver tetradrachm of Bagadat (Baydad), priest-king of Persis (modern south-west Iran), late third or early second century BC. The idea of placing the ruler's portrait on the coin and its style are derived from Greek practices introduced after Alexander the Great, although the appearance of the subject is Iranian, with a moustache and earring. (× 3)

**44g** Silver tetradrachm of Demetrius I, King of Bactria (modern Afghanistan), early second century BC. The Greek king is shown wearing an elephant scalp headdress. This had also been used for some portraits of Alexander the Great, and symbolised the domination of India, which, according to Greek legend, had been conquered by the Greek god Dionysus in an elephant chariot.

**44h** Gold aureus of Augustus, Roman emperor 31 BC–AD 14. The portrayal of individuals on Roman coinage during their lifetime was introduced by Julius Caesar in 44 BC and became a feature of Roman imperial coins. The Roman emperors copied this idea from Hellenistic practice, although they avoided the symbols of kingship such as the diadem and instead portrayed themselves wearing a laurel wreath, a symbol of victory.

**44i** The Hellenistic kingdoms *c.* 275 BC.



**45** Silver tetradrachm of Philip II, King of Macedon, 359–336 BC. Under Philip, the Macedonian kingdom began to expand, and he conquered all the independent cities in northern Greece and took possession of the gold and silver mines there. This access to wealth was an important factor in Philip's rise to power, and he minted large numbers of coins. The silver coins depict a young rider on a huge racehorse and commemorate Philip's own victory at the Olympic Games of 356 BC. The Greek inscription gives the king's name (*Philippou*). The rider is shown holding a laurel branch and wearing a wreath, both symbols of victory. (× 3)

large quantities. Evidence for the widespread use of coinage in day-to-day transactions in the late fifth century is found in the plays of Aristophanes, where characters talk of buying fish and sickles in the market-place, and of storing small coins in their mouths! And the vignettes of the fourth-century satirist Theophrastus contain numerous references to coins – buying, selling or lending money. By this stage the Athenians had started to mint low-denomination bronze coins for everyday use: excavations in the Athenian Agora have brought to light over 16,000 coins, mostly of bronze, which had presumably been dropped. The sheer number of these finds vividly illustrates the extent to which coined money had become the norm by this time. Similarly, most hoards dating from the end of the Classical period consist of coins rather than, as had earlier been the case, of bullion. Monetised societies depended upon coinage for most financial transactions, and as a consequence states were required to maintain the reputation and value of their coins in various ways. At Athens the passing of fake coins was strictly forbidden, and a law of 375 BC instituted the checking of coins by public slaves.

**46** Silver tetradrachm in the name of Alexander the Great (336–323 BC), struck in 280 BC at Cyzicus in Turkey. Alexander made large numbers of gold and silver coins to finance his conquests. The coins, struck at many mints throughout his empire, were of standard types; the silver depicted Heracles and the seated figure of Zeus. The same designs continued long after Alexander's death.

## Conclusion

Access to gold and silver was one of the secrets behind the extraordinary success of the Macedonians under Philip II (359–336 BC) and his son Alexander the Great (336–323 BC) in conquering Greece and defeating the might of the wealthy Persian Empire. Philip's seizure of the mines of Thrace enabled him to produce a massive coinage in gold, silver and bronze to pay his army, while Alexander's conquest of Asia gave him control of the accumulated wealth of the Persian king – estimated at 180,000 talents in total – all of which was subsequently distributed, either as coin or booty. These are sums of a completely different order from those we have met thus far, and foreshadow the imperial wealth of the Roman Empire after its conquest of the Mediterranean world. Coinage was produced in greater quantities and over a much wider area in the Hellenistic period, but the basic monetary culture did not change.

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# 2 The Roman World

*Around [the Mediterranean] lie the continents far and wide, pouring an endless flow of goods to you [Rome]. There is brought from every land and sea whatever is brought forth by the seasons and is produced by all countries, rivers, lakes, and the skills of Greeks and foreigners. . . . So many merchantmen arrive here with cargoes from all over, at every season, and with each return of the harvest, that the city seems like a common warehouse of the world.*

These words, taken from the speech *To Rome* written by Aelius Aristides, a second-century AD Greek rhetorician from Asia Minor, convey something of the enormous scale of the city of Rome as the metropolis of a great empire. Rome had grown to a city of perhaps over a million inhabitants, the largest European city known to history until the eighteenth century. With this growth of city and empire came an economy and use of money on a vast scale, altogether unprecedented in the ancient world of the Mediterranean and the Near East.

## Early Rome

Rome itself had already had a long history before Aelius Aristides' time, having been founded, according to Roman tradition, in 753 BC. Rome's origins lay in a small city, at first governed by kings and then by a republican constitution under an aristocratic ruling class, and for the first 400 or so years of the city's history it developed within a strictly regional and Italian context. One aspect of Rome's geographical and cultural limits was the lack of coined money, a situation that endured until about 300 BC, some 200 years after it had become a normal feature of life in the Greek world of the Aegean and indeed of the Greek colonies in Sicily and southern Italy.

Later writers thought that the earliest form of money in Rome had been sheep and cattle, as they derived the Latin word for 'money', *pecunia*, from the word for 'cattle', *pecus*. But, while it is possible that certain values had been expressed in terms of head of cattle in Rome as elsewhere in the ancient world, it is less likely that they ever served as a form of payment. There is better evidence for the monetary use of bronze bullion measured by weight among the early Romans, as well as among the other indigenous peoples of Italy, including the Etruscans. Since bronze is a relatively cheap metal, considerable quantities were needed: the historian Livy (first century BC) mentions a tradition that Roman senators from the time of the early Republic had to carry their wealth in carts (Livy, 4.60.6). From this period date the early roughcast lumps of